

JOSEPH SRUJAN EDA

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PROFESSIONAL SUMMARY

Embedded Software Engineer specializing in **C/C++**, **Embedded Linux (Yocto)**, **field experience in oil field installations**, and **IoT Gateways**. Expertise in building multi-threaded applications, kernel customization for **ARM (AM335x)**, and industrial protocols (**Modbus**, **MQTT**). Proven track record in field deployment and system reliability optimization.

TECHNICAL SKILLS

- **Oil Field & Asset Monitoring:** Modbus, Device & RTU Configurations, SCADA Management, RTU Assembly, Panel Board Wiring, Firmware Upgrades, System Troubleshooting.
- **Languages & OS:** C, Embedded C, C++, Python, Shell (Bash), Linux, RTOS, Yocto Project (BitBake).
- **Protocols & HW:** Modbus RTU/TCP, MQTT, UART, SPI, I2C, TCP/IP, STM32, AM335x, LPC2148.
- **Tools & Concepts:** Git, GCC, GDB, Keil, VS Code, Multi-threading, IPC, Socket Programming, DSA.

PROFESSIONAL EXPERIENCE

Embedded Developer | Kellton Tech, Hyderabad — April 2024 – Present

- **Field Deployment (Oil India):** Represented Kellton at **Oil India** client sites; successfully installed and commissioned IoT gateway panels in active oil fields, ensuring hardware-software synergy in rugged environments.
- **Gateway Configuration App:** Designed a Windows desktop application with an intuitive GUI for configuring IoT gateways, eliminating manual setup and reducing commissioning effort by 80%.
- **Yocto & Kernel Optimization:** Customized Yocto layers and kernel parameters; **reduced build time by 20%** and improved connection reliability by **30%**.
- **HMI Display:** Interfaced and programmed a UART industrial HMI display for the IoT gateway, developing custom UI pages for real-time parameter visualization, and local field diagnostics.
- **Industrial Automation:** Integrated **Modbus RTU/TCP** for real-time monitoring and remote VSD control; built a Windows GUI tool that reduced setup effort by **80%**.

Advanced Embedded Systems Trainee | Vector India, Hyderabad — Aug 2023 – Feb 2024

- Developed bare-metal drivers for **ARM & 8051** (UART, I2C, SPI); mastered Linux IPC and synchronization.

EDUCATION

- Bachelor of Technology in Electronics and Communication Engineering | *Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool* | 2019 – 2023
- Class XII | *Tirumala Junior College / Andhra Pradesh Board of Intermediate Education* | 2019
- Class X (Secondary) *Sri Chaitanya School / Board of Secondary Education, Andhra Pradesh* | 2017

TECHNICAL PROJECTS

- **Multi-Purpose E-Card (C, ARM LPC2148):** Developed an RFID-based smart card system with SPI-based drivers for peripheral interfacing and unified identity verification.
- **Smart Gate Keeping (Python, OpenCV):** Automated vehicle monitoring using OCR and CNNs; achieved 98% recognition accuracy with automated CSV logging.

CERTIFICATIONS & ACHIEVEMENTS

- HackerRank C (5-Star), C++ (4-Star) | NPTEL Gold Elite (90%) - Python.
- **Internship:** AI/ML Intern @ Samsung Innovation Campus (Data preprocessing & Neural Networks).
- Leetcode [150+ Problems].
- **Operations Industrial Engineer Job Simulation** certification from **SIEMENS**.
- **Data Science Essentials with Python** certification from **CISCO**.
- **Databases for Developers** certification from **ORACLE**.